MILITARY SPECIFICATION

REGULATOR, VOLTAGE: 10539715

1. SCOPE

1.1 This specification covers one type of transistorized direct current voltage regulator. This regulator delivers a regulated 16 ± 0.2 volt direct current output when energized by an external 18 volt to 30 volts direct current power source.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18, Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

C10539538	Schematic 16 volt Regulator and Reticle Lamp Control
D10539715	Regulator, Voltage: 10539715

FSC 1240

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Packaging Data Sheet

MIL-P-14232/P10539715 Packaging of Voltage Regulator

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Initial production sample.- Requirements for the submission of an initial production sample by the contractor shall be as specified in the contract.

3.2 Fabrication.- The regulator shall be manufactured in accordance with Drawing D10539715 and drawings pertaining thereto.

3.2.1 Moisture-fungus resistance.- The regulator shall be treated per Type II for moisture and fungus resistance in accordance with MIL-T-152.

3.3 General specifications.- The regulator shall meet the following requirements of MIL-F-13926 and MIL-C-46357. Should any conflict exist between the general specifications and this specification, the order of precedence shall be: this specification, MIL-C-46357, and MIL-F-13926.

3.3.1 MIL-C-46357.

(a) Etched circuit assembly
(b) Marking
(c) Handling

3.3.2 MIL-F-13926.

(a) Dimensions and tolerances
(b) Inorganic protective surface finishes
(c) Workmanship

3.4 Environmental service conditions.

3.4.1 Vibration.- The regulator shall withstand a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion), at a frequency varying uniformly between 10 and 55 cycles per second (cps) each minute, for a total of 10 minutes duration in each of 3 mutually perpendicular axes (total of 30 minutes) relative to its normal operating position. The regulator shall not be damaged or adversely affected by vibration and shall subsequently meet the requirements of 3.5.

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3.4.2 Shock .- The regulator shall withstand 5 shock impulses of 15 gravitational Units (G's). Time duration of each impulse shall be 11 ± 1 millisecond. The regulator shall not be damaged or adversely affected by shock and shall subsequently meet the requirements of 3.5.

3.4.3 Temperatures.

3.4.3.1 operating.- The regulator shall meet the requirements of 3.5.2 and 3.5.4 when thermally stabilized at temperatures ranging from -400 to +150° Fahrenheit (F).

3.4.3.2 Storage.- The regulator shall meet the requirements of 3.5 at a temperature of between 70° and 80° F after being thermally stabilized at each of the extreme temperatures of -65° and +160°F.

3.4.4 Humidity.

3.4.4.1 Operating.- The regulator shall meet the requirements of 3.5.2 and 3.5.4 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of +155°F.

3.4.4.2 Non-operating. - The regulator shall meet the requirements of 3.5 (when thoroughly dried) after exposure to an atmosphere with a RH of 100%, including condensation for a period of 4 hours.

3.4.5 Salt spray.- The regulator shall show no evidence of physical damage and shall meet the requirements of 3.5 after being exposed to an atomized spray of a 5% salt solution concentration.

3.5 Performance.

3.5.1 Input voltage.- The regulator shall be energized by an input voltage of from 18 to 30 volts direct current (vdc).

3.5.2 Output voltage.- The regulator shall deliver a regulated output voltage of 16 ± 0.2 vdc, with load conditions between zero and 700 milliamperes (ma).

3.5.3 Input voltage surge.- There shall be no change in the output voltage as specified in 3.5.2 after the regulator is subjected to five 500 ± 25 milliseconds dc voltage surges of 50 volts each at the rate of one surge per minute. Subsequent to the input surges, the regulator shall show no evidence of physical or electrical damage.

3,5.4 Rise time.- The regulator shall meet the output voltage requirements of 3.5.2 within 100 milliseconds of applying an input voltage specified in 3.5.1.

3.5.5 Regulation.

3.5.5.1 Regulation relative to input voltage variations.- Regulation relative to the input voltage variation shall be 2% or less for any input voltage change as specified in 3.5.1, under two constant load conditions of one-half to full load.

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3.5.5.2 Regulation relative to output load variations.- Regulation relative to the output load variation shall be 2% or less for a varying load from one-half to full load at any constant input voltage within the limits specified in 3.5.1.

3.5.6 Electrical shock.- The regulator under maximum output load conditions shall show no evidence of damage when subjected to one hour of intermittent operation consisting of a continuously repetitive 30 seconds "ON" and 30 seconds "OFF" cycle of input voltage application.

3.5.7 Reverse Polarity- The regulator shall not be damaged by the application of an operational input potential of reverse polarity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 and the component and subassembly inspection requirements for etched circuit assembly, marking and handling of MIL-C-46357 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be treed in regular production. Initial production testing shall be in accordance with 4.3, 4.4, and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.1 Initial production sample failure.- Should one unit fail to meet the initial production testing specified in 4.2, the contractor shall cease those operations which are causing the rejection and make the necessary corrections thereto. When corrections in the process or controls have been made, two additional (new) production models shall be submitted in accordance with 4.2.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Tables I, II, and III herein. Examinations and tests for Packaging, Packing, and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLAss	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR:- AQL 1.0% Defective			
101.	Vibration	3.4.1	4.6.2
102.	Performance of electrical tests	3.5	4.6.9
103.	Input voltage	3.5.1	4.6.9.1
104.	Electrical shock	3.5.6	4.6.9.2
105.	Reverse polarity	3.5.7	4.6.9.3
106.	Output voltage	3.5.2	4.6.9.4
107.	Input voltage surge	3.5.3	4.6.9.5
108.	Rise time	3.5.4	4.6.9.6
109.	Output regulation relative to input voltage variations	3.5.5.1	4.6.9.7
110.	Output regulation relative to output load variations	3.5.5.2	4.6.9.8
MINOR: NONE			

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One regulator shall be selected at random as a special sample from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURES
301.	Fabrication	3.2	Applicable drawings - visual
302.	General specification	3.3.1	MIL-C-46357-visual and tactile
303.	General specification	3.3.2	MIL-F-13926-visual and tactile

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4.4.2 Environmental.- Three regulators shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

TABLE III

	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURES
304.	Shock (non-operating)	3.4.2	4.6.3
305.	Storage temperatures (non-operating)	3.4.3.2	4.6.4
306.	Humidity (non-operating)	3.4.4.2	4.6.5
307.	Salt-spray (non-operating)	3.4.5	4.6.6
308.	Temperature (operating)	3.4.3.1	4.6.7
309.	Humidity (operating)	3.4.4.1	4.6.8

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall suply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with the design requirements of 4.5.2.1, the following standard test equipment shall be utilized in the performance of the applicable tests specified in 4.6.

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ITEM	NOMENCLATURE	DESCRIPTION
1	Regulated Voltage Supply	INPUT : 105 vac to 135 vac at 45 cps to 480 CPS OUTPUT : a. Voltage: Continuously variable, zero to 40 vdc b. Regulation: 0.015% or 1.0 millivolt (Line to load) c. Ripple: 250 microvolts(rms), 1.0 millivolt (p-p) max. d. Current: Zero to 1.5 dc amperes at 71°C(Centigrade) e. Temperature coefficient: 0.015%/°C
2	Voltmeter, DC	a. Range: A range of at least 0.001 to 30.0 vdc b. Input impedance: No less than ten megohms on all ranges c. Accuracy: At least 0.01% when reading ± 1 count

4.6 Test methods and procedures.

4.6.1 Moisture-fungus resistance non-operating condition.- One of the first five units produced shall be submitted for inspection after treatment (see 3.2.1) to determine compliance with inspection provisions of MIL-T-152 and then subjected to the fungus test defined in MIL-F-13926 to determine compliance with requirement 3.2.1. After approval, subsequent acceptance for moisture-fungus resistance may be made by certification.

4.6.2 Vibration non-operating condition.- This test shall utilize standard vibrating equipment with means for mounting the regulator, measuring and controlling direction of vibration, and for adjusting the amplitude and frequencies to comply with the limits specified in 3.4.1. Test equipment measuring instruments employed shall measure the amplitude of vibration within ± 5% and the frequency of vibration within ± 2%. Vibration shall be as specified in 3.4.1 respective to regulator normal operational position and under conditions specified in the "Test facilities" requirement of MIL-F-13926. The regulator shall subsequently be inspected to determine compliance with the performance requirements of 3.5.

4.6.3 Shock non-operating condition.- This test shall be made under conditions specified in the "Test facilities" requirement of MIL-F-13926 and in accordance with conditions specified in 3.4.2 and Procedure 11 of the shock test defined in MIL-F-13926, except that the number of impact shocks shall be as specified in 3.4.2 and shall be applied perpendicular to the board's mounting surface only. The regulator shall subsequently be inspected to determine compliance with the performance requirements of 3.5.

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4.6.4 Storage temperatures non-operating condition.- This test shall utilize test equipment conforming with the "Test facilities requirement of MIL-F-13926. Test facilities shall maintain conditions specified in 3.4.3.2.Temperatures shall be gradually changed in the performance of this test as a precautionary measure to avoid the possibility of thermal shock. After regulator temperature stabilization at -65°F and +160°F, the temperature shall be maintained constant for 24 hours at each temperature specified. The temperature shall then be returned to between 70°F and 80°F, at which time the regulator shall be inspected for compliance with performance requirements of 3.5.

4.6.5 Humidity non-operating condition.- This test shall be conducted in accordance with the humidity test defined in Procedure 111 of MIL-F-13926, except that the relative humidity shall be the value specified in 3.4.4.2 for the prescribed time period and the test shall cease at the end of that period. Upon return to a temperature of between 60°F and 90°F the external surfaces of the regulator shall be thoroughly dried by wiping all accessible areas. Drying by air blast or in an oven shall not be permitted. After a period of not more than two hours the regulator shall be inspected at a temperature of between 70°F and 80°F to determine compliance with the requirements of 3.5.

4.6.6 Salt spray non-opeating condition.- This test shall be performed in accordance with the conditions specified in requirement 3.4.5 and the salt spray test defined in MIL-STD-202, Condition B. The regulator shall be given a visual examination for physical damage and subsequently inspected to determine compliance with the requirements of 3.5.

4.6.7 Temperature-operating. - This test shall utilize test equipment specified in 4.6.9 in addition to environmental test equipment conforming with the "Test facilities" requirement of MIL-F-13926. Test facilities shall maintain conditions specified herein in accordance with requirement 3.4.3.1. Temperatures shall be gradually changed in the performance of this test as a precautionary measure to avoid the possibility of thermal shock. After the regulator has attained thermal stabilization at -40°F and then at +150°F, temperature shall be maintained constant at each temperature while determining compliance with requirement 3.5.2 and 3.5.4 and for no less than one hour of operation at each specified temperature. This test shall be conducted after the pretest conditions specified in 4.6.9 have been met between 70°F and 80°F ,

4.6.8 Humidity-operating. - This test shall utilize test equipment specified in 4.6.9 and shall be conducted in accordance with the humidity test defined in Procedure III of MIL-F-13926, except that temperature and humidity shall be maintained at the value specified in requirement 3.4.4,1 for a 24 hour period and the test shall cease at the end of that period. Under the above conditions, the regulator shall be tested for compliance with requirement 3.5.2 and 3.5.4 in accordance with requirement 3.4.4.1 only after the pretest conditions specified in 4.6.9 have been met at 70°F to 80°F.

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4.6.9 Performance of electrical tests.- Tests herein shall utilize as applicable an external power source (power supply) and dc voltmeter (to verify regulator output voltage) having the characteristics of Item 1 and 2 respectively in 4.5.3. Also utilized, as applicable, shall be a vacuum tube voltmeter (VTVM, to verify regulator input voltage and polarity), a non-inductive variable resistive load (regulator output load simulator), and a dc milliammeter (to verify output load dc). Polarities on Drawing C10539538 shall be observed nless otherwise specified. Electrical tests shall be made under a maximum relative humidity of 80%, a barometric pressure of 28 to 30 inches of mercury, at between 70°F and 80°F except tests 4.6.9.4 and 4.6.9.6, which shall be under conditions specified in 3.4.3.1 and 3.4.4.1. A regulator pretest shall be made with a 24 vdc input to determine that regulator output voltage demonstrates operational stability as specified herein. Preset regulator output in accordance with the following pretest procedure: Employing test equipment specified herein, preset Regulator Output Voltage potentiometer (R6), Drawing CI0539538, to obtain a regulator output of +16 vdc, with the regulator output load adjusted to obtain a regulator output load current of 500 dc milliamperes (ma). After one hour of continuous operation an output of +16 ± .05 vdc shall indicate acceptable operational stability. Reset R6 to +16 vdc as necessary to compensate for any drift and conduct subsequent electrical tests. Inability of the regulator to demonstrate the operational stability specified herein shall be considered an output voltage test failure (see 4.6.9.4) and shall be cause for rejection.

4.6.9.1 Input voltage.- This test shall employ the power source and VTVM specified in 4.6.9 to verify that regulator input voltage is of the polarity specified on Drawing CI0539538 and that a variation of input voltage from 18 vdc to 30 vdc will permit the regulator to remain operational without evidence of physical or electrical damage.

4.6.9.2 Electrical shock.- This test shall utilize test equipment specified in 4.6.9 in addition to a manual or automatic electrical switch and a standard time measuring device. A regulator input of 30 vdc shall be switched ON and OFF in accordance with the time period specified in 3.5.6, with the regulator output load adjusted to obtain a regulator output dc of 700 ma. This test shall be conducted after test 4.6.9.1 and prior to conducting the remaining electrical tests specified herein.

4.6.9.3 Reverse polarity.- This test shall utilize test equipment specified in 4.6.9 in addition to a standard time measuring device and shall be conducted after test 4.6.9.2. Apply a regulator input of 30 vdc in opposite polarity to that indicated on Drawing c10539538 for one minute and observe that the regulator shows no evidence of damage in compliance with requirement 3.5.7.

4.6.9.4 Output voltage.- This test shall utilize test equipment specified in 4.6.9. To determine compliance with requirement 3.5.2, apply a regulator input of 30 vdc and adjust regulator output load resistance

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to obtain a regulator dc output of 700 ma. Observe that the regulator output voltage is +16 ± 0.2 vdc with respect to regulator ground, which shall be negative. Regulator output voltage shall remain +16 ± 0.2 vdc as the regulator output load resistance is increased until less than 10 dc ma, of flow exists through the regulator output load. The regulator output voltage shall remain +16 ± 0.2 vdc after disconnecting the regulator output load from the regulator output circuit.

4.6.9.5 Input voltage surge.- This test shall utilize test equipment specified in 4.6.9 in addition to an oscilloscope, to verify the amplitude and time duration of each voltage surge specified in requirement 3.5.3, and a means of generating a positive going voltage pulse of from zero vdc to +50 vdc. This rise and decay time of this pulse is not critical but the overall pulse shape shall approximate a square wave. A suggested method of generating this pulse is shown in Figure 1. To determine compliance with requirement 3.5.3 apply a regulator input of 30 vdc, adjust the regulator output load to obtain a regulator dc output of 700 ma. Simultaneously, remove regulator input voltage and immediately apply the input voltage surge specified in 3.5.3. Immediately after passage of each surge a regulator input voltage of 30 vdc shall be applied and maintained until the next surge is applied. Observe that the regulator output voltage level remains +16 ± 0.2 vdc in compliance with requirement 3.5.3 after passage of the specified surges and that no damage to the regulator resulted therefrom.

4.6.9.6 Rise time.- This test shall utilize test equipment specified in 4.6.9 in addition to a standard measuring device. With a regulator imput of 18 vdc, adjust the regulator output load to obtain a regulator dc output of 700 ma. To determine compliance with requirement 3.5.4, observe that the regulator output voltage specified in requirement 3.5.2 appears within 100 milliseconds from the instant of input voltage application.

Regulation relative to input voltage variation.- The test equipment specified in 4.6.9 shall be employed for this test. Adjust the regulator output load to obtain a regulator dc output of 700 ma (full-load output). Observe that the regulator demonstrates compliance with 3.5.5.1, in accordance with the formula specified herein, when the regulator input voltage is varied from vdc to 30 vdc. Adjust the regulator output load to obtain a regulator dc output of 350 ma (half-load output). Observe that the regulator demonstrates compliance with requirement 3.5.5.1 when the regulator input voltage is varied from 18 vdc to 30 vdc.

$$\text{Percent Regulation} = \frac{E_{max} - E_{min}}{E_{min}} \times 100$$

When:

$E_{max.}$ = Regulator output vdc with vdc regulator input

$E_{min.}$ = Regulator output vdc with 18 vdc regulator input

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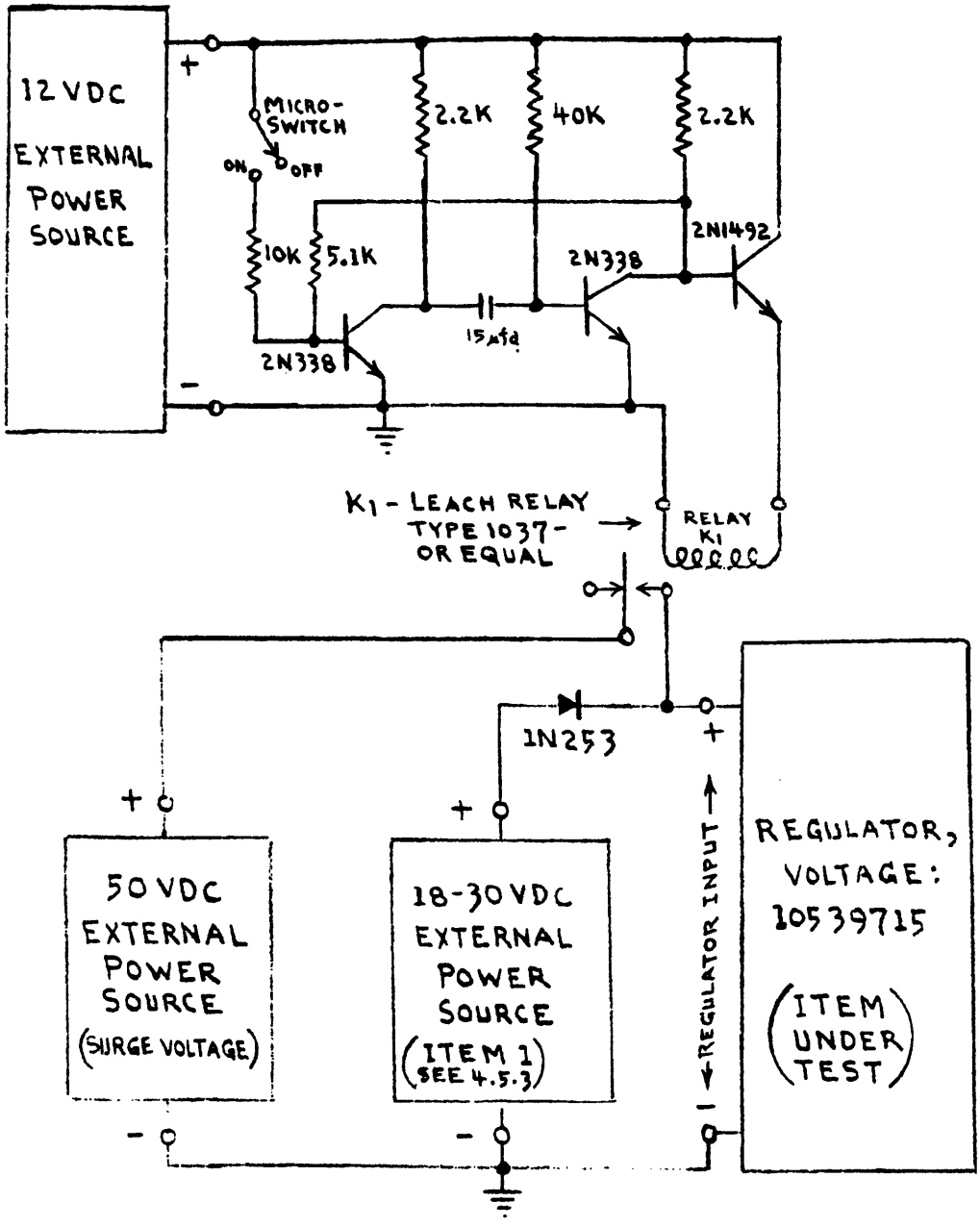


FIGURE 1. SUGGESTED PULSE GENERATOR

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